



MAKHTESHIM
CHEMICAL WORKS LTD.

SAFETY DATA SHEET

EU Regulation No. 1907/2006, Annex II

Page : 1 of 7

Revised edition no : 6

Date : 5 / 12 / 2012

Supersedes : 13 / 4 / 2011

Novaluron 10% EC

R-9075.DPD

SECTION 1. Identification of the substance/mixture and of the company/undertaking

1.1. Product identifier

Trade name : Novaluron 10% EC
Chemical formula : Novaluron : C₁₇H₉ClF₈N₂O₄
Novaluron : 1-[3-chloro-4-(1,1,2-trifluoro-2-trifluoro-methoxyethoxy)phenyl]- 3-(2,6-difluorobenzoyl)urea

1.2. Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses : Insecticide
Uses advised against : None under normal conditions.

1.3. Details of the supplier of the safety data sheet

: Makhteshim Chemical Works Ltd.
PO Box 60
Beer Sheva 84100 Israel.
Fax : +972-8-6296848
Email : msds@mcw.co.il

1.4. Emergency telephone number

Phone nr : 972-3-6106666 Subscription no. 36789 (Makhteshim)

SECTION 2. Hazards identification

2.1. Classification of the substance or mixture

Classification 67/548/EEC (DSD) - 1999/45/EC (DPD)

: Repr. Cat. 2; R61
Xi; R36/37/38
R43
N; R50-53

2.2. Label elements

Labelling 1999/45/EC (DPD)

Symbol(s)



R Phrase(s)

S Phrase(s)

: T : Toxic
N : Dangerous for the environment.
: R36/37/38 - Irritating to eyes, respiratory system and skin.
R43 - May cause sensitization by skin contact.
R50/53 - Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.
R61 - May cause harm to the unborn child.
: S02 - Keep out of the reach of children.
S20/21 - When using, do not eat, drink or smoke.
S24/25 - Avoid contact with skin and eyes.
S26 - In case of contact with eyes, rinse immediately with plenty of water and seek medical advice.
S28 - After contact with skin, wash immediately with plenty of
S36/37/39 - Wear suitable protective clothing, gloves and eye/face protection.
S61 - Avoid release to the environment. Refer to special instructions/Safety data sheets.
water

2.3. Other hazards

Other hazards : None under normal conditions.

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SECTION 2. Hazards identification (continued)

SECTION 3. Composition/information on ingredients

Components

: This product is considered to be hazardous and contains hazardous components.

Substance name	Concentration % w/w	CAS No	EC No	Index No	REACH Reg.	Classification
Novaluron	100 g/l	116714-46-6	-----	-----	----	N; R50-53 Aquatic Acute 1 - H400 Aquatic Chronic 1 - H410
N-methyl-2-pyrrolidone	ca 450 g/l	872-50-4	212-828-1	606-021-00-7	----	Repr. Cat. 2; R61 Xi; R36/37/38 Repr. 1B - H360D Eye Irrit. 2 - H319 Skin Irrit. 2 - H315 STOT SE 3 - H335
Phosphate esters emulsifier	165 g/l	-----	-----	-----	----	Xi; R36/38 Eye irrit. 2 - H319 Skin Irrit. 2 - H315

Text of R/H-Phrases - Sec. 16

SECTION 4. First aid measures

4.1. Description of first aid measures

First aid measures

Inhalation

: Remove victim to fresh air. Consult a doctor in the event of any complaints.

Skin contact

: Remove affected clothing and wash all exposed skin area with soap and water, followed by warm water rinse. Consult a doctor in the event of any complaints.

Eye contact

: Wash off with plenty of water (15 min.) If pain persists, consult an eye specialist.

Ingestion

: Do not induce vomiting. Wash out mouth with plenty of water.
Never give anything by mouth to an unconscious person. Consult a doctor in the event of any complaints.

4.2. Most important symptoms and effects, both acute and delayed

4.3. Indication of any immediate medical attention and special treatment needed

Note to physician

: No specific antidote known.
Treat symptomatically and give supportive therapy.
If ingested perform gastric lavage and administer activated charcoal.

SECTION 5. Fire-fighting measures

5.1. Extinguishing media

Suitable extinguishing media

: For small fire : dry chemical , carbon dioxide .
For large fire: foam , water fog , water spray .

5.2. Special hazards arising from the substance or mixture

Specific hazards

: Thermal decomposition may generate : carbon dioxide , carbon monoxide , nitrogen oxides , chlorides , fluorides .

5.3. Advice for fire-fighters

Protection against fire

: Wear suitable protective clothing. Self-contained breathing apparatus.

Special procedures

: Fight fires from a protected location . Dike fire control water for later disposal.

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SECTION 6. Accidental release measures

6.1. Personal precautions, protective equipment and emergency procedures

For emergency responders : Use appropriate protection (see section 8).

6.2. Environmental precautions

Environmental precautions : Dispose of this material and its container at hazardous or special waste collection point. If the product has contaminated surface water, inform the appropriate authorities. Contaminated soil layers have to be dug out.

6.3. Methods and material for containment and cleaning up

Clean up methods : In the event of minor spillage: Absorb in sand or other inert material . Use appropriate containment to avoid environmental contamination.
In the event of major spillage: Dike spills using absorbent or impervious materials such as sand or clay for later disposal. Collect and contain as much free liquid as possible. Do not empty into drains .

6.4. Reference to other sections

SECTION 7. Handling and storage

7.1. Precautions for safe handling

Handling : Do not breathe vapour . Avoid contact with skin and eyes.

7.2. Conditions for safe storage, including any incompatibilities

Storage : Keep locked up . Keep container tightly closed. Keep only in the original container in a cool, well ventilated place.

7.3. Specific end use(s)

SECTION 8. Exposure controls/personal protection

8.1. Control parameters

TLV® -TWA [mg/m³] : (USA) Novaluron : Not established
(USA) N-Methyl-2-pyrrolidone : Not established

8.2. Exposure controls

- Respiratory protection : During spraying wear suitable respiratory equipment.
- Hand protection : Wash hands thoroughly after handling.
Facilities storing or utilizing this material should be equipped with an eyewash facility and safety shower . Wash clothing before re-using.
- Skin protection : Wear suitable gloves.
- Eye protection : Wear suitable protective clothing & chemical resistant boots .
Chemical goggles or safety glasses.

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SECTION 9. Physical and chemical properties

9.1. Information on basic physical and chemical properties

Appearance	: Liquid
Colour	: light yellow
Odour	: Faint characteristic
Odour threshold	: Not available
pH	: Not available
Melting point / Freezing point	
Melting point [°C]	: Novaluron : 176.5 - 178°C (Pure)
Initial boiling point - boiling range	: Not available
Flash point	: 102°C (Closed cup)
Evaporation rate	: Not available
Flammability	: Not flammable
Explosion limits (lower - upper)	: Not available
Vapour pressure	: Novaluron : 1.6x(10) ⁻⁵ Pa at a temperature of 25°C (Pure)
Vapour density	: Not available
Relative density	: 1.07
Solubility	: Novaluron : 3 µg/l at a temperature of 25°C
Partition coefficient : n-octanol / water	: Novaluron : 4.3
Auto-ignition temperature	: Not available
Decomposition temperature	: Not available
Viscosity	: Not available
Explosive Properties	: Not available
Oxidising properties	: Not available

9.2. Other information

: Not available

SECTION 10. Stability and reactivity

10.1. Reactivity

10.2. Chemical stability

Chemical stability : Stable under normal conditions.

10.3. Possibility of hazardous reactions

Hazardous reactions : Hazardous polymerization will not occur .

10.4. Conditions to avoid

Conditions to avoid : Protect from sunlight , open flame and sources of heat .

10.5. Incompatible materials

Materials to avoid : Avoid contact with : strong acids , strong bases .

10.6. Hazardous decomposition products

Hazardous decomposition products : Thermal decomposition may generate : carbon dioxide , carbon monoxide , nitrogen oxides , chlorides , fluorides .

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SECTION 11. Toxicological information

11.1. Information on toxicological effects

- Acute toxicity
 - Oral, Rat LD50 (mg/kg) : > 5000
 - Dermal, Rat LD50 [mg/kg] : > 2000
 - Inhalation, Rat LC50 [mg/l/4h] : Novaluron Tech: > 5.15
- Skin corrosion/irritation : Slight skin irritant.
- Serious eye damage/irritation : Irritating to eyes.
- Respiratory or skin sensitisation : Guinea pig maximization test: Sensitizer . However, the field-use dilution (0.1% in distilled water) does not cause skin sensitization
- Germ cell mutagenicity : No evidence of genotoxicity
- Carcinogenicity : Not carcinogenic
- Reproductive toxicity : N-methyl-2-pyrrolidone: Repro. Cat. 2 May cause harm to the unborn child
- STOT-single exposure : Not available
- STOT-repeated exposure : Not available
- Aspiration hazard : Not available

SECTION 12. Ecological information

12.1. Toxicity

Acute toxicity

- 96 H-LC50 - Rainbow trout [mg/l] : 62.4
- 48 H-EC50 - Daphnia magna [µg/l] : 4.22
- 72 H-EbC50 - Algae [mg/l] : 33.5
- 72 H-ErC50 - Algae [mg/l] : 87.2
- LD50 Birds [mg/kg] : > 2000

Long-term exposure

- 48 H-NOEC - Daphnia magna [µg/l] : 1.25

12.2. Persistence - degradability

- Persistence - degradability : Novaluron : Half-life time (t1/2) : 68-76 days , soil

12.3. Bioaccumulative potential

12.4. Mobility in soil

12.5. Results of PBT and vPvB assessment

- : Not available

12.6. Other adverse effects

- : None known.

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SECTION 13. Disposal considerations

13.1. Waste treatment methods

General

: Dispose in a safe manner in accordance with local/national regulations.

SECTION 14. Transport information

14.1. UN number

UN N°

: 3082

Hazard Label(s)



14.2. UN proper shipping name

- Land transport - ADR/RID : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (100 g/l Novaluron)
- Sea transport - IMO-IMDG : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (100 g/l Novaluron)
- Air transport - ICAO/IATA : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (100 g/l Novaluron)
- Proper shipping name : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.

14.3. Transport hazard class(es)

- Land transport - ADR/RID : 9
- Sea transport - IMO-IMDG : 9
- IMO-IMDG - class or division : 9
- Air transport - ICAO/IATA : 9

14.4. Packing group

- Land transport - ADR/RID : III
- Sea transport - IMO-IMDG : III
- IMO-IMDG - Packing group : III
- Air transport - ICAO/IATA : III

14.5. Environmental hazards

- IMDG-Marine pollution : Marine pollutant - Polluant marin

14.6. Special precautions for user

: See transport documents.

14.7. Transport in bulk according to Annex II of MARPOL73/78 and the IBC Code

: Not applicable.

SECTION 15. Regulatory information

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

: Ensure all national/local regulations are observed.

15.2. Chemical Safety Assessment

: Not applicable

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SECTION 16. Other information

List of relevant R phrases (heading 3) : *R36/37/38 - Irritating to eyes, respiratory system and skin.*
R36/38 - Irritating to eyes and skin.
R50/53 - Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.
R61 - May cause harm to the unborn child.

List of relevant H phrases (Section 3) : *H315 - Causes skin irritation.*
H319 - Causes serious eye irritation.
H335 - May cause respiratory irritation.
H360D - May damage the unborn child.
H400 - Very toxic to aquatic life.
H410 - Very toxic to aquatic life with long lasting effects.

Print date : 8 / 9 / 2013

Recommended uses and restrictions : None.

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